

PROF. DR.-ING. MARK SCHUTERA

# DGX SPARK

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These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at [https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes\\_dgxspark](https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_dgxspark). Every contribution is welcome.



2026-10-01 · local draft





*“POCA FAVILLA GRAN FIAMMA SECONDA.”*

*“A GREAT FLAME FOLLOWS A LITTLE SPARK.”*

*DANTE ALIGHIERI, PARADISO I, 34*

# Contents

*Quickstart.*      9

*Index*      10

# *Introduction*

THIS IS A MANUAL. It gets you onto the lab's NVIDIA DGX Spark: A desktop machine with one GB10 Grace Blackwell chip and 128 GB of memory shared between CPU and GPU, running DGX OS, NVIDIA's Ubuntu. It is here for training runs, not as an inference server. This is a powerful compute substrate, use it. Then again you will probably never be alone on this machine, so be kind to data, jobs, and artifacts on this machine. Sandbox whatever you do, so you don't hamper the work of others. On the other hand, expect that someone else could nuke your stuff, so backups are advised. If you put stuff on that machine, expect that someone else will be able to see and use it.

This is currently a quickstart only, so chapters on how to setup a proper project structure and environment is highly appreciated. Same holds for a chapter on how to effectively run jobs on a shared GPU. Anything is worth adding, that you would have loved to have known before diving deep. The goal is to have a short, concise, and practical manual for the lab's DGX Spark over time. If a step is deprecated or you found a better way, contribute it back.

THE OFFICIAL DOCUMENTATION.  
NVIDIA's user guide and the playbooks  
at <https://build.nvidia.com/spark>.



# Quickstart.

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## Ignition

THE SPARK HAS ONE USER ACCOUNT PER PERSON. Request yours by mail. The reply carries your username and password, and the setup folder as an attached zip with the WireGuard configuration and the Spark's IP address.

YOUR ACCOUNT IS AN ADMIN ACCOUNT on a shared machine. With sudo you can install what you need, and also break the machine for everyone. With great power comes great responsibility: Store nothing sensitive on it, leave other users' files and processes alone, sandbox your stuff and never touch the system config itself. Protect yourself from others, and protect others from yourself. While you have your own user, two users being logged in at the same time will produce lag and slowdowns. If you experience this, please log out and try again later. That also means the Spark is used most efficiently when you spend as little time as possible to deploy your jobs and leave, not for interactive development. If you need to develop interactively, please do so on your own machine.

WIREGUARD OPENS A TUNNEL INTO THE LAB NETWORK. Only traffic to the lab goes through it; the rest of your connection stays as it is. Install the WireGuard client, choose Import tunnel(s) from file, pick dgx-spark.conf from the setup folder, and press Activate. The status should now read Active and Latest handshake is seconds old. No handshake means your network blocks the tunnel.

THEN LOG IN with the IP address from the mail:

```
ssh <username>@<spark-ip>
```

BEFORE YOU START A JOB, run `nvidia-smi` to see who is on the GPU.

THIS SETUP IS FAR FROM PERFECT. If you want to build a job scheduler, per-user VPN configurations, or sandboxes that keep users apart, reach out.

SCAN THE CODE and send a mail stating your name, why you need access, which course, and what you want to run.



REMEMBER? Chapter 1 of Full Stack Handwerkszeug, Development and Production, covers this split, along with SSH and Linux servers. [https://material.schutera.com/lecturenotes/notes\\_missingsemester/notes\\_missingsemester.pdf](https://material.schutera.com/lecturenotes/notes_missingsemester/notes_missingsemester.pdf)

YOU LEARN USEFUL STUFF HERE. QED.

DOWNLOAD. <https://www.wireguard.com/install/>

# *Index*

account, [9](#)

DGX Spark, [9](#)

license, [2](#)

nvidia-smi, [9](#)

quickstart, [9](#)

shared infrastructure, [9](#)

SSH, [9](#)

sudo, [9](#)

WireGuard, [9](#)